Northwind SQL Practice Questions

Parts A — with Solutions and Explanations

Part A — Foundational SELECT Queries

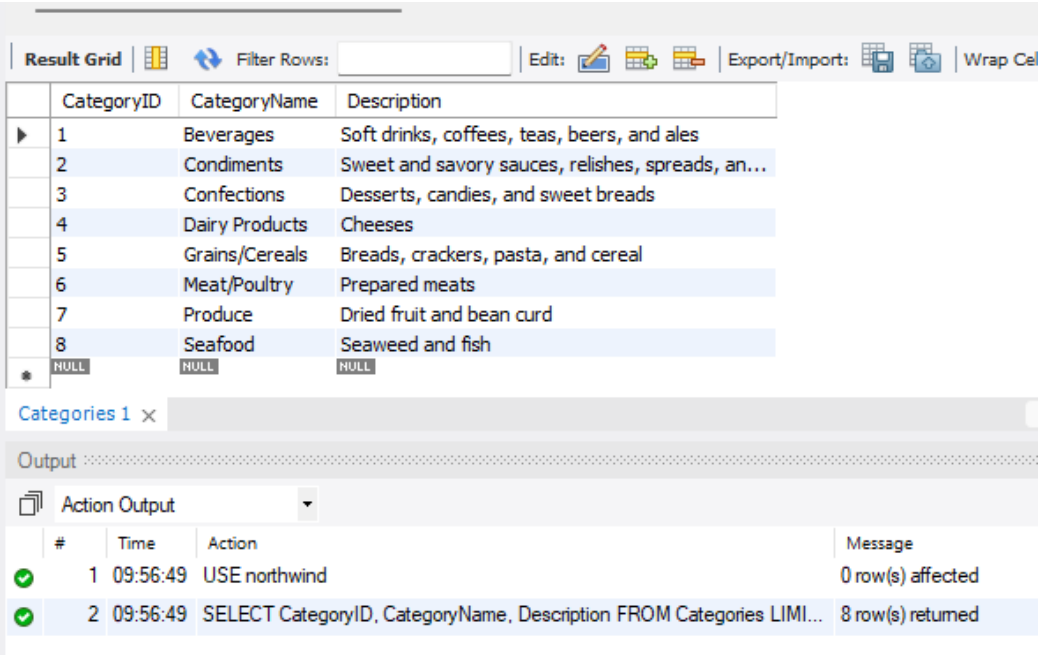
The questions in this section focus on core SELECT syntax: choosing columns, filtering rows with WHERE, using comparison and pattern-matching operators, and writing simple aggregate queries. All queries use the Northwind sample database.

Question 1

Task: Write a query to list the CategoryID, CategoryName, and Description of every category in the Northwind database.

Hint: Look at the Categories table. Use SELECT to pick only the columns you need.

Expected Output



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cel

CategoryID	CategoryName	Description
1	Beverages	Soft drinks, coffees, teas, beers, and ales
2	Condiments	Sweet and savory sauces, relishes, spreads, an...
3	Confections	Desserts, candies, and sweet breads
4	Dairy Products	Cheeses
5	Grains/Cereals	Breads, crackers, pasta, and cereal
6	Meat/Poultry	Prepared meats
7	Produce	Dried fruit and bean curd
8	Seafood	Seaweed and fish
NULL	NULL	NULL

Categories 1

Output

Action Output

#	Time	Action	Message
1	09:56:49	USE northwind	0 row(s) affected
2	09:56:49	SELECT CategoryID, CategoryName, Description FROM Categories LIMI...	8 row(s) returned

Question 2

Task: Write a query to find all products where the quantity per unit contains the word 'box' (e.g., '12 boxes' or '24 - 12 oz boxes').

Hint: Use the LIKE operator with wildcard characters to search for partial text matches inside the QuantityPerUnit column of the Products table.

Expected Output

Result Grid | Filter Rows: | Edit: | Export/Import:

ProductID	ProductName	QuantityPerUnit
1	Chai	10 boxes x 20 bags
5	Chef Anton's Gumbo Mix	36 boxes
13	Konbu	2 kg box
16	Pavlova	32 - 500 g boxes
19	Teatime Chocolate Biscuits	10 boxes x 12 pieces
20	Sir Rodney's Marmalade	30 gift boxes
47	Zaanse koeken	10 - 4 oz boxes
52	Filo Mix	16 - 2 kg boxes
55	Pt chinois	24 boxes x 2 pies
68	Scottish Longbreads	10 boxes x 8 pieces
77	Original Frankfurter grne ...	12 boxes
NULL	NULL	NULL

Question 3

Task: Write a query to list all products that are currently discontinued. Display the ProductID, ProductName, and Discontinued columns.

Hint: The Discontinued column in the Products table stores 1 for discontinued and 0 for active.

Expected Output

Result Grid | Filter Rows: | Edit: | Export/Import:

ProductID	ProductName	Discontinued
5	Chef Anton's Gumbo Mix	1
9	Mishi Kobe Niku	1
17	Alice Mutton	1
24	Guaran Fantstica	1
28	Rssle Sauerkraut	1
29	Thringer Rostbratwurst	1
42	Singaporean Hokkien Fried Mee	1
53	Perth Pasties	1
NULL	NULL	NULL

Question 4

Task: Write a query to display the full name of every employee by combining their first and last name into a single column. Label the combined column FullName.

Hint: Look at the Employees table. Use string concatenation (+ in SQL Server, || in SQLite/PostgreSQL) to combine FirstName and LastName with a space between them.

Expected Output

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EmployeeID	FullName	Title
1	0	Sales Representative
2	0	Vice President, Sales
3	0	Sales Representative
4	0	Sales Representative
5	0	Sales Manager
6	0	Sales Representative
7	0	Sales Representative
8	0	Inside Sales Coordinator
9	0	Sales Representative

Result 4

Output

Action Output

#	Time	Action	Message
1	10:09:22	SELECT EmployeeID, FirstName + ' ' + LastName AS FullName, ...	9 row(s) returned

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Question 5

Task: Write a query to list all customers located in either Germany or France. Display the CustomerID, CompanyName, and Country.

Hint: You can use IN to test against a list of values instead of writing multiple OR conditions.

Expected Output

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Conte

CustomerID	CompanyName	Country
ALFKI	Alfreds Futterkiste	Germany
BLAUS	Blauer See Delikatessen	Germany
BLONP	Blondesddsl pre et fils	France
BONAP	Bon app'	France
DRACD	Drachenblut Delikatessen	Germany
DUMON	Du monde entier	France
FOLIG	Folies gourmandes	France
FRANK	Frankenversand	Germany
FRANR	France restauration	France
KOENE	Kniglich Essen	Germany
LACOR	La corne d'abondance	France
LAMAI	La maison d'Asie	France
LEHMS	Lehmanns Marktstand	Germany

Customers 6

Output

Action Output

#	Time	Action	Message
1	10:12:29	SELECT CustomerID, CompanyName, Country FROM Customers WHER...	22 row(s) returned

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Question 6

Task: How many total orders have been placed in the Orders table? Write a query that returns this count and labels it TotalOrders.

Hint: Use the COUNT() aggregate function on the primary key of the Orders table.

Expected Output

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

TotalOrders
830

Result 7

Output

Action Output

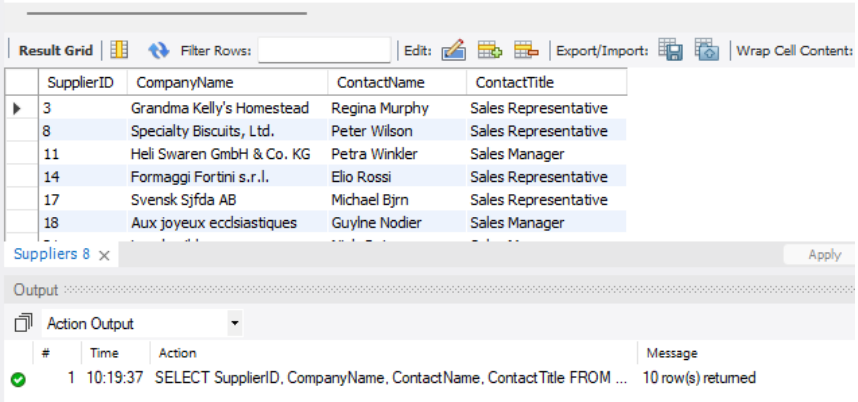
#	Time	Action	Message
1	10:15:50	SELECT COUNT(OrderID) AS TotalOrders FROM Orders LIMIT 0, 1000	1 row(s) returned

Question 7

Task: Write a query to list all suppliers whose contact title is either 'Sales Representative' or 'Sales Manager'. Show SupplierID, CompanyName, ContactName, and ContactTitle.

Hint: Use the IN operator or two OR conditions in your WHERE clause against the Suppliers table.

Expected Output



SupplierID	CompanyName	ContactName	ContactTitle
3	Grandma Kelly's Homestead	Regina Murphy	Sales Representative
8	Specialty Biscuits, Ltd.	Peter Wilson	Sales Representative
11	Heli Swaren GmbH & Co. KG	Petra Winkler	Sales Manager
14	Formaggi Fortini s.r.l.	Elio Rossi	Sales Representative
17	Svensk Sjfda AB	Michael Bjrn	Sales Representative
18	Aux joyeux ecclsiastiques	Guylne Nodier	Sales Manager

Suppliers 8

Output

Action Output

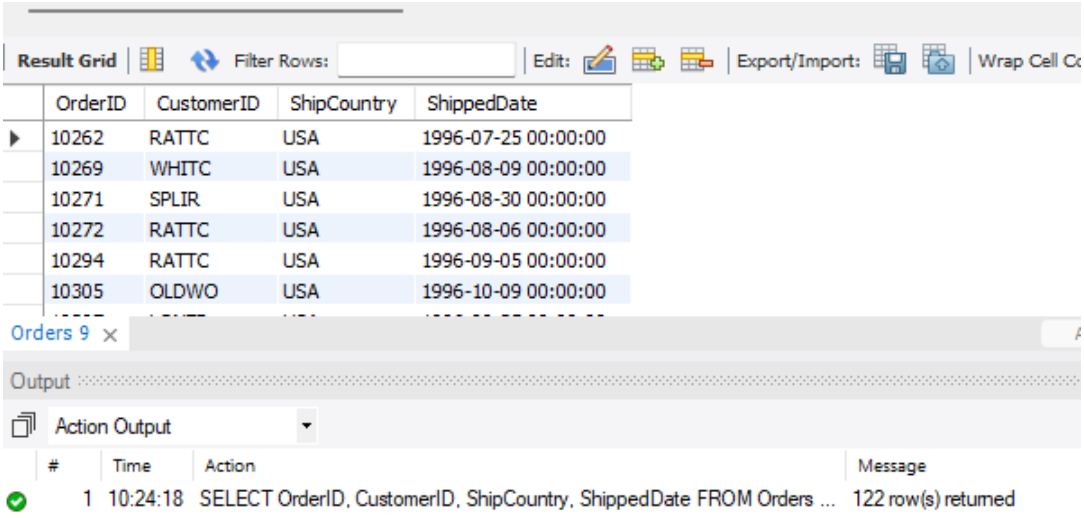
#	Time	Action	Message
1	10:19:37	SELECT SupplierID, CompanyName, ContactName, ContactTitle FROM ...	10 row(s) returned

Question 8

Task: Write a query to find all orders that were shipped to the USA. Display OrderID, CustomerID, ShipCountry, and ShippedDate.

Hint: Filter the Orders table using the ShipCountry column.

Expected Output



OrderID	CustomerID	ShipCountry	ShippedDate
10262	RATTC	USA	1996-07-25 00:00:00
10269	WHITC	USA	1996-08-09 00:00:00
10271	SPLIR	USA	1996-08-30 00:00:00
10272	RATTC	USA	1996-08-06 00:00:00
10294	RATTC	USA	1996-09-05 00:00:00
10305	OLDWO	USA	1996-10-09 00:00:00

Orders 9

Output

Action Output

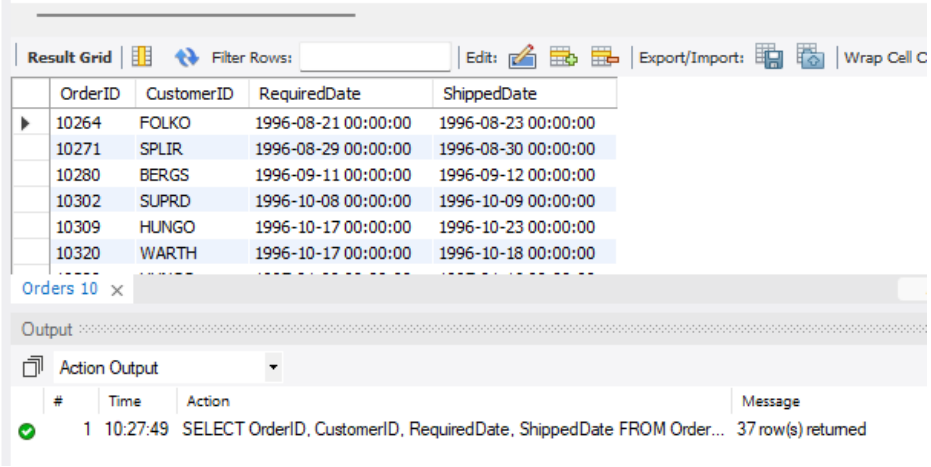
#	Time	Action	Message
1	10:24:18	SELECT OrderID, CustomerID, ShipCountry, ShippedDate FROM Orders ...	122 row(s) returned

Question 9

Task: Examine the Orders table. It records a required date and an actual shipped date. Write a query to find all orders where the product was shipped after the required date (i.e., the shipment was late). Display OrderID, CustomerID, RequiredDate, and ShippedDate.

Hint: Compare two date columns using the > operator in a WHERE clause.

Expected Output



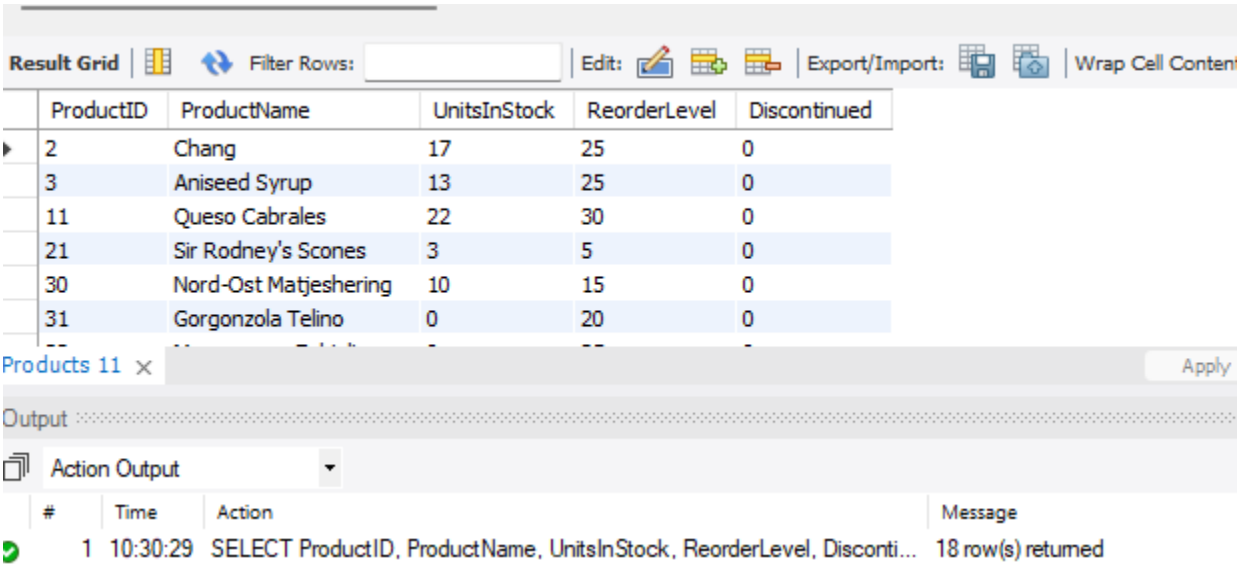
OrderID	CustomerID	RequiredDate	ShippedDate
10264	FOLKO	1996-08-21 00:00:00	1996-08-23 00:00:00
10271	SPLIR	1996-08-29 00:00:00	1996-08-30 00:00:00
10280	BERGS	1996-09-11 00:00:00	1996-09-12 00:00:00
10302	SUPRD	1996-10-08 00:00:00	1996-10-09 00:00:00
10309	HUNGO	1996-10-17 00:00:00	1996-10-23 00:00:00
10320	WARTH	1996-10-17 00:00:00	1996-10-18 00:00:00

Question 10

Task: Examine the Products table. It contains a column called UnitsInStock and a column called ReorderLevel. Write a query that identifies products that need to be reordered — that is, products where the current stock is at or below the reorder level and the product is NOT discontinued.

Hint: Combine two conditions in the WHERE clause using AND.

Expected Output



ProductID	ProductName	UnitsInStock	ReorderLevel	Discontinued
2	Chang	17	25	0
3	Aniseed Syrup	13	25	0
11	Queso Cabrales	22	30	0
21	Sir Rodney's Scones	3	5	0
30	Nord-Ost Matjeshering	10	15	0
31	Gorgonzola Telino	0	20	0

Part A Solutions

Question 1

```sql
USE northwind;

--  Part A Question 1
SELECT CategoryID, CategoryName, Description
FROM Categories;
```

Explanation

- SELECT specifies the columns to display: CategoryID, CategoryName, and Description.
- FROM Categories tells SQL which table to retrieve the data from.
- No WHERE clause is needed because we want every row in the table.
- This is a foundational SELECT statement — columns are comma-separated and the statement ends with a semicolon.

Question 2

```sql
-- Part A Question 2
SELECT ProductID, ProductName, QuantityPerUnit
FROM Products
WHERE QuantityPerUnit LIKE '%box%';
```

Explanation

- LIKE is used for pattern matching in text columns.
- The % wildcard matches any sequence of characters (including none) on either side of the word 'box'.
- Using '%box%' finds values containing 'box' anywhere in the string, whether it appears at the start, middle, or end.
- This is case-insensitive in most SQL dialects (SQL Server, MySQL).

Question 3

```sql
-- Part A Question 3
SELECT ProductID, ProductName, Discontinued
FROM Products
WHERE Discontinued = 1;
```

Explanation

- In the Northwind database the Discontinued column is a bit/boolean field.
- A value of 1 means the product has been discontinued; 0 means it is still active.
- The WHERE clause filters rows to show only discontinued products.
- This demonstrates filtering on a boolean/bit column — a very common pattern in database applications.

Question 4:

```sql
-- Part A Question 4
SELECT EmployeeID,
       FirstName + ' ' + LastName AS FullName,
       Title
FROM Employees;
```

Explanation

- The + operator in SQL Server (or || in other dialects) joins strings together.
- ' ' (a single space between two single quotes) adds a space between the first and last name.
- AS FullName gives the combined expression a readable column alias.
- Aliases are applied with the AS keyword and appear as the column header in the result set.

Question 5

```sql
-- Part A Question 5
SELECT CustomerID, CompanyName, Country
FROM Customers
WHERE Country IN ('Germany', 'France');
```

Explanation

- The IN operator is a shorthand for multiple OR equality checks.
- WHERE Country IN ('Germany', 'France') is equivalent to WHERE Country = 'Germany' OR Country = 'France'.
- String values must be enclosed in single quotes.
- IN keeps the query readable when checking against several possible values.

Question 6

```sql
SELECT COUNT(OrderID) AS TotalOrders
FROM Orders;
```

Explanation

- COUNT() is an aggregate function that counts the number of non-NULL values in a column.
- Counting on the primary key (OrderID) guarantees that every row is counted.
- AS TotalOrders assigns a meaningful alias to the result column.
- Aggregate functions collapse multiple rows into a single summary value — here, one row with the total.

Question 7

```sql
-- Part A Question 7
SELECT SupplierID, CompanyName, ContactName, ContactTitle
FROM Suppliers
WHERE ContactTitle IN ('Sales Representative', 'Sales Manager');
```

Explanation

- This query targets the Suppliers table using IN for a clean multi-value filter.
- The IN list contains string literals, each enclosed in single quotes and separated by commas.
- This approach is easy to extend — just add more values inside the parentheses.
- Compared to multiple OR conditions, IN is typically more readable and equally efficient.

Question 8

```sql
-- Part A Question 8
SELECT OrderID, CustomerID, ShipCountry, ShippedDate
FROM Orders
WHERE ShipCountry = 'USA';
```

Explanation

- This is a straightforward equality filter on a text column.
- String comparisons in SQL are enclosed in single quotes.
- The ShippedDate column will show NULL for orders not yet shipped — a real-world consideration.
- Column selection (not SELECT *) is a good practice: it makes the query's intent clear and improves performance on large tables.

Question 9

```sql
-- Part A Question 9
SELECT OrderID, CustomerID, RequiredDate, ShippedDate
FROM Orders
WHERE ShippedDate > RequiredDate;
```

Explanation

• You can compare two columns directly in a WHERE clause — you are not limited to comparing a column against a fixed value.

• ShippedDate > RequiredDate returns only rows where the shipped date is later (greater) than the required date, meaning the order was late.

• Rows where ShippedDate IS NULL (order not yet shipped) are automatically excluded because comparisons with NULL always evaluate to unknown/false in SQL.

Question 10

```sql
-- Part A Question 10
SELECT ProductID, ProductName, UnitsInStock, ReorderLevel, Discontinued
FROM Products
WHERE UnitsInStock <= ReorderLevel
  AND Discontinued = 0;
```

Explanation

• <= means 'less than or equal to', so the filter catches products at or below their reorder threshold.

• AND requires BOTH conditions to be true for a row to appear in results.

• Adding AND Discontinued = 0 avoids flagging products that are no longer sold — a practical real-world requirement.

• This query demonstrates combining numeric comparisons with boolean/bit column filters.